Lyapunov decay and zero-dissipation orbits in damped AVF systems

Abstract

We study the distinction between physical-energy dissipation and strict Lyapunov control for partially damped average-vector-field systems. A fixed-step example has a globally coercive nonincreasing energy, an explicit locally strict quadratic energy, and a cold periodic orbit of zero numerical dissipation, although its continuous dynamics converge globally. We then construct a different single real-analytic, globally strongly convex potential with bounded Hessian. Its implicit map is unique at every positive step, and one explicit local quadratic energy is strict for all steps in $(0, 1]$. Nevertheless, a sequence of steps tending to zero admits exact cold periodic orbits whose initial energies tend to infinity. Thus no one small-step threshold guarantees convergence from every phase for that potential. No failure on a fixed compact family of starts is asserted. Both constructions retain the numerical invariant-set qualifications in the established passivity theory.

Keywords: Lyapunov function; discrete gradient; partial damping; observability; invariant set

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1 Introduction

A physical energy often proves that every trajectory stays bounded, although its derivative vanishes on a set larger than the equilibrium. Convergence then depends on the trajectories which remain in that set. A time discretization may preserve the energy identity while changing this invariant-set question. It is therefore useful to distinguish preservation of energy dissipation from preservation of the asymptotic behavior.

Discrete-gradient methods provide exact energy identities for a large class of dissipative systems. Celledoni and Høiseth [1] formulate this property for port-Hamiltonian systems and derive convergence under a separate numerical zero-state observability hypothesis. The average-vector-field method is one of their examples. A discrete invariant-set hypothesis also appears in the more recent control formulation of Macchelli [2]. Related energy-balance constructions for differential-algebraic systems are developed by Kinon, Morandin and Schulze [3]. These results explain why the invariant set must be checked for the numerical map itself.

We consider one fixed smooth potential. An explicit quadratic function decreases along the actual nonlinear numerical map in a specified neighborhood, but an exact orbit outside it prevents globally strict state-energy decay. The method, energy identity, quadratic Lyapunov equation and local perturbation argument are standard. This is a qualified verification for that map; no firstness is asserted. Classical time-discrete observability

results such as [4] concern linear conservative systems with their own spectral and mesh conditions.

The first construction occupies Sections 1–4. Sections 5–8 give a separate potential for which the obstruction occurs along steps tending to zero, while a common local strict energy remains valid. Its witnesses have initial energies tending to infinity. This separates an all-phase step threshold from one allowed to depend on a compact set of starts. General nearby-attractor and semiglobal results [5, 6] have their own method and uniformity assumptions, which are not silently transferred to the exact nonpolynomial AVF integral.

Write $q = (x, y) \in \mathbb{R}^2$, $v = (u, w) \in \mathbb{R}^2$ and $Z = (x, y, u, w) \in \mathbb{R}^4$. Fix

$$U(x, y) = \frac{x^2}{2} + 2y^2 + \frac{1}{4}\sin x \sin y, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{1}$$

and define the physical energy

$$H(q, v) = U(q) + \frac{1}{2}\|v\|^2. \tag{2}$$

The continuous system is

$$\dot{q} = v, \qquad \dot{v} = -\nabla U(q) - Dv. \tag{3}$$

In Sections 1–4 the discrete step is exactly $h = 1$. The average-vector-field equations are

$$\begin{aligned} q^+ - q &= \frac{v^+ + v}{2}, \\ v^+ - v &= -\int_0^1 \nabla U\big(q + t(q^+ - q)\big)\, dt - D\frac{v^+ + v}{2}. \end{aligned} \tag{4}$$

The integral is exact. We shall prove that these equations define a unique smooth map $Z^+ = \Phi Z$ on all of $\mathbb{R}^4$.

For the local energy, in the stated coordinate order, put

$$P = \frac{1}{4032}\begin{pmatrix} 34375 & 325458 & 7600 & -20194 \\ 325458 & 4255335 & 80766 & -2520 \\ 7600 & 80766 & 12640 & 60572 \\ -20194 & -2520 & 60572 & 1061560 \end{pmatrix}. \tag{5}$$

Define

$$T = \operatorname{tr} P = \frac{297995}{224}, \qquad \ell = \frac{\det P}{T^3}, \qquad r^2 = \frac{1}{400T}, \qquad \eta = \frac{\ell r^2}{2}, \tag{6}$$

and $V(Z) = Z^T P Z$.

Theorem 1. *For the potential* (1) *and the discrete system* (4)*, the following statements hold.*

(i) *The potential is real analytic and*

$$\frac{3}{4}I \preceq \nabla^2 U(q) \preceq \frac{17}{4}I \quad (q \in \mathbb{R}^2). \tag{7}$$

Its unique minimizer is the origin. Every implicit step has a unique solution, and Φ *is smooth on* $\mathbb{R}^4$*. For every phase,*

$$\frac{3}{8}\|Z\|^2 \le H(Z) \le \frac{17}{8}\|Z\|^2, \tag{8}$$

$$H(\Phi Z) - H(Z) = -\left(\frac{u^+ + u}{2}\right)^2. \tag{9}$$

Thus every forward orbit exists and satisfies

$$\|Z_n\|^2 \le \frac{8}{3}H(Z_0), \qquad \sum_{n=0}^{\infty}\left(\frac{u_{n+1}+u_n}{2}\right)^2 \le H(Z_0). \tag{10}$$

(ii) *The matrix* P *is positive definite,* $\ell > 0$*, and*

$$\ell\|Z\|^2 \le V(Z) \le T\|Z\|^2. \tag{11}$$

For every $\|Z\| \le r$*, the actual nonlinear map obeys*

$$V(\Phi Z) - V(Z) \le -\frac{1}{2}\|Z\|^2. \tag{12}$$

The sublevel $\{V \le \eta\}$ *is positively invariant. For every* Z_0 *in that sublevel and every* $n \ge 0$*,*

$$V(Z_n) \le \left(1 - \frac{1}{2T}\right)^n V(Z_0), \qquad \|Z_n\|^2 \le \ell^{-1}\left(1 - \frac{1}{2T}\right)^n V(Z_0). \tag{13}$$

(iii) *The origin of* (3) *is globally asymptotically stable.*

(iv) *With* $A = 2\pi$*, the four phases*

$$\begin{aligned} Z_0 &= (0, A, 0, 0), \quad & Z_1 &= (0, 0, 0, -2A), \\ Z_2 &= (0, -A, 0, 0), \quad & Z_3 &= (0, 0, 0, 2A) \end{aligned} \tag{14}$$

form an exact prime four-cycle of Φ*, in the displayed order. This orbit starts with zero velocity, has constant energy* $8\pi^2$ *and zero numerical output* $(u_{n+1} + u_n)/2$*, and has precisely these four points as its omega-limit set. For every fixed positive integer* m*, there is no real-valued function* W *on* $\mathbb{R}^4$ *such that*

$$W(\Phi^m Z) < W(Z) \qquad \text{for every } Z \ne 0. \tag{15}$$

The global capture in part (i) is proved before the local estimates in part (ii). The theorem does not assert attraction to the four-cycle, robustness of that cycle, or failure of all local Lyapunov functions. In particular, part (ii) gives a nonempty neighborhood whose trajectories do converge. All statements concern the single step and single potential fixed above.

2 The physical energy and the implicit map

Lemma 1. *The map Φ is globally well defined and smooth, and part* (i) *of Theorem 1 holds.*

Proof. The Hessian of the trigonometric term before multiplication by $1/4$ is

$$B(x,y) = \begin{pmatrix} -\sin x \sin y & \cos x \cos y \\ \cos x \cos y & -\sin x \sin y \end{pmatrix}.$$

Its eigenvalues are $\cos(x+y)$ and $-\cos(x-y)$. Hence $-I \preceq B \preceq I$, and adding $\operatorname{diag}(1,4)$ proves (7). Since $U(0)=0$ and $\nabla U(0)=0$, Taylor's formula gives

$$\frac{3}{8}\|q\|^2 \le U(q) \le \frac{17}{8}\|q\|^2. \tag{16}$$

Strong convexity makes the origin the unique minimizer and the only critical point. Adding the kinetic energy proves (8).

For a fixed arbitrary phase, let $a = q^+ - q$. The first equation of (4) forces $v^+ = 2a - v$, and the second is equivalent to

$$(2I + D)a + \int_0^1 \nabla U(q + ta)\, dt = 2v. \tag{17}$$

This is the critical-point equation for

$$J_{q,v}(a) = \|a\|^2 + \frac{1}{2}a^T D a - 2\langle v, a\rangle + \int_0^1 \frac{U(q+ta) - U(q)}{t}\, dt. \tag{18}$$

The integrand at $t=0$ has the limiting value $\langle \nabla U(q), a\rangle$. Its smooth extension also follows by writing that quotient as $\int_0^1 \langle \nabla U(q + sta), a\rangle\, ds$. Differentiation under the integral is valid on compact parameter sets, and

$$\nabla_a^2 J_{q,v}(a) = 2I + D + \int_0^1 t\nabla^2 U(q+ta)\, dt \succeq \frac{19}{8} I. \tag{19}$$

Consequently $J_{q,v}$ is coercive and strongly convex. It attains its unique minimum, which solves (17). This proves existence and uniqueness for the original equations at every phase. The derivative in (19) is everywhere invertible. The implicit-function theorem gives local smooth solution functions, and uniqueness makes them agree, yielding a global smooth map.

For this actual solution put $\bar v = (v^+ + v)/2 = a$ and $\bar g = \int_0^1 \nabla U(q+ta)\, dt$. The chain rule along the segment and the kinetic difference give

$$\begin{aligned} H(q^+, v^+) - H(q, v) &= \langle \bar g, a\rangle + \langle \bar v, v^+ - v\rangle \\ &= \langle \bar v, \bar g + v^+ - v\rangle = -\bar v^T D \bar v = -\left(\frac{u^+ + u}{2}\right)^2. \end{aligned} \tag{20}$$

This proves (9). Its iteration and the lower bound in (8) prove capture. Summing the same identity over a finite interval and using $H \geq 0$ bounds the nonnegative partial sums in (10); their limit gives the asserted series bound. ∎

This argument is the usual discrete-gradient energy argument, supplemented here by a global proof of the implicit solve. The dissipative output is an average over a step. Its vanishing does not say that either endpoint velocity is zero, nor does the series estimate control every phase increment.

3 An explicit local strict energy

We now verify the local function on the actual nonlinear map. Put

$$\begin{aligned} K &= \begin{pmatrix} 1 & 1/4 \\ 1/4 & 4 \end{pmatrix}, \qquad L_0 = \begin{pmatrix} 0 & I \\ -K & -D \end{pmatrix}, \\ G &= (I - L_0/2)^{-1}, \qquad M = G(I + L_0/2). \end{aligned} \tag{21}$$

Direct rational arithmetic gives

$$M = \frac{1}{895} \begin{pmatrix} 641 & -32 & 512 & -16 \\ -48 & 1 & -16 & 448 \\ -508 & -64 & 129 & -32 \\ -96 & -1788 & -32 & 1 \end{pmatrix}, \qquad M^T P M - P = -I. \tag{22}$$

The leading principal minors of the integer numerator $4032P$ are

$$\begin{aligned} 34375, \qquad & 40354230861, \\ 439602459331140, \qquad & 284781267819132789600. \end{aligned} \tag{23}$$

They are positive, so P is positive definite. Also

$$\begin{aligned} \det P &= \frac{35590967375525}{33030144}, \qquad T = \frac{297995}{224} > 1, \\ \|G\|_F^2 &= \frac{447528}{160205} < 9, \qquad \|M\|_F^2 = \frac{872916}{160205} < 9. \end{aligned} \tag{24}$$

If the eigenvalues of P are $0 < \lambda_1 \leq \cdots \leq \lambda_4$, each is at most T and $\det P \leq \lambda_1 T^3$. Therefore $\ell \leq \lambda_1$ and (11) follows. These exact identities can also be reproduced by the accompanying verification program.

Lemma 2. *For every $Z \in \mathbb{R}^4$, the actual map satisfies*

$$\Phi Z = MZ + N(Z), \qquad \|N(Z)\| \leq 11\|Z\|^3. \tag{25}$$

Proof. Write $\nabla U(q) = Kq + R(q)$. Then

$$R(x, y) = \frac{1}{4} \begin{pmatrix} \cos x \sin y - y \\ \sin x \cos y - x \end{pmatrix}.$$

The inequalities $|\sin t - t| \le |t|^3/6$ and $|1 - \cos t| \le t^2/2$ give

$$|R_1(x,y)| \le \frac{1}{8}|y|(x^2+y^2), \qquad |R_2(x,y)| \le \frac{1}{8}|x|(x^2+y^2).$$

Squaring and adding yields the global bound

$$\|R(q)\| \le \frac{1}{8}\|q\|^3. \tag{26}$$

The physical-energy estimate, already proved independently of locality, gives

$$\|q^+\|^2 \le \frac{8}{3}H(\Phi Z) \le \frac{8}{3}H(Z) \le \frac{17}{3}\|Z\|^2. \tag{27}$$

Thus every point $q_t = (1-t)q + tq^+$ has norm at most $3\|Z\|$. Splitting the linear terms in (4) gives the exact identity

$$(I - L_0/2)\Phi Z = (I + L_0/2)Z - \begin{pmatrix} 0 \\ \displaystyle\int_0^1 R(q_t)\,dt \end{pmatrix}. \tag{28}$$

Use (26) and $\|G\| < 3$ from (24) to obtain

$$\|\Phi Z - MZ\| \le 3\int_0^1 \frac{(3\|Z\|)^3}{8}\,dt = \frac{81}{8}\|Z\|^3 \le 11\|Z\|^3.$$

This proves the claim without assuming that the successor stays in a local ball. ∎

Proof of Theorem 1(ii). Set $s = \|Z\|$. By (22), Lemma 2 and $\|P\| \le T$, $\|M\| < 3$,

$$\begin{aligned} V(\Phi Z) - V(Z) &= -s^2 + 2(MZ)^T PN(Z) + N(Z)^T PN(Z) \\ &\le -s^2 + 66Ts^4 + 121Ts^6. \end{aligned} \tag{29}$$

If $s \le r$, then $s^2 \le 1/(400T) < 1$, and hence

$$66Ts^2 + 121Ts^4 \le 187Ts^2 \le \frac{187}{400} < \frac{1}{2}.$$

Substitution in (29) proves (12), including $Z = 0$.

For $V(Z) \le \eta$, coercivity gives $\|Z\|^2 \le \eta/\ell = r^2/2$. The just-proved decrement therefore applies and yields $V(\Phi Z) \le V(Z) \le \eta$. This proves invariance before it is used to iterate the estimate. Within this sublevel,

$$V(\Phi Z) \le V(Z) - \frac{1}{2}\|Z\|^2 \le \left(1 - \frac{1}{2T}\right)V(Z).$$

Here the factor is strictly between zero and one. Iteration and (11) give (13). ∎

4 Continuous convergence and the exact orbit

Proof of Theorem 1(iii). The Hessian bound makes the vector field in (3) globally Lipschitz, so every initial phase has a global forward solution. Along it,

$$\frac{d}{dt}H(q(t),v(t)) = -u(t)^2. \tag{30}$$

The coercive bounds (8) give compact capture and $\|Z(t)\| \le \sqrt{17/3}\,\|Z(0)\|$. In particular the origin is Lyapunov stable.

Suppose a trajectory stays in $u = 0$. Then $x = c$ is constant, and its first acceleration equation reads

$$c + \frac{1}{4}\cos c \sin y(t) = 0. \tag{31}$$

If $\cos c = 0$, this would force $c = 0$, a contradiction. Otherwise $\sin y(t)$ is constant. A level set of sine is discrete, including at the values ± 1; the continuous image of a time interval is connected. Thus y is constant, and $w = 0$. Both acceleration equations now imply $\nabla U(c, y) = 0$, so strong convexity gives $(c, y) = 0$. The origin is the only trajectory that remains in the zero-output set.

For completeness, let a bounded forward trajectory have energy limit H_∞, and let a be any of its omega-limit points. Choose $t_j \to \infty$ with $Z(t_j) \to a$. For every fixed $s \ge 0$, continuous dependence of the autonomous flow gives $\varphi_s(a) = \lim_j Z(t_j + s)$, and therefore $H(\varphi_s(a)) = H_\infty$. By (30), the forward trajectory from a stays in $u = 0$, so $a = 0$. All omega-limit points are zero. Compactness then implies $Z(t) \to 0$, proving global asymptotic stability. ∎

Proof of Theorem 1(iv). For each proposed transition in (14), both endpoint values of x and u are zero. Denote the old and new second positions by a and b. Since $a, b \in \{0, A, -A\}$ and $a \ne b$ on each edge,

$$\int_0^1 \sin\bigl(a + t(b-a)\bigr)\,dt = \frac{\cos a - \cos b}{b - a} = 0. \tag{32}$$

The first averaged force consequently vanishes. The second averaged force is $2(a + b)$, since $\sin x = 0$ along the whole segment. The remaining equations reduce to

$$b - a = \frac{w^+ + w}{2}, \qquad w^+ - w = -2(a + b). \tag{33}$$

The following exact values verify both identities on every edge:

edge	$b - a$	$(w^+ + w)/2$	$w^+ - w$	$-2(a + b)$
$0 \to 1$	$-A$	$-A$	$-2A$	$-2A$
$1 \to 2$	$-A$	$-A$	$2A$	$2A$
$2 \to 3$	A	A	$2A$	$2A$
$3 \to 0$	A	A	$-2A$	$-2A$

(34)

The zero first-coordinate equations hold as well. Thus all four successors solve the literal implicit equations. Their global uniqueness, proved in Lemma 1, identifies them as the actual map values.

The four phases are distinct, the initial velocity is zero, and each has $H = 2y^2+w^2/2 = 2A^2 = 8\pi^2$. Every midpoint output is zero. Each of these phases recurs infinitely often, and no other phase occurs; the omega-limit set is therefore exactly the four-point set in (14).

Fix $m \geq 1$. Every phase $\Phi^{jm}Z_0$ belongs to that cycle and is nonzero, and $\Phi^{4m}Z_0 = Z_0$. Applying (15) four times would give

$$W(Z_0) > W(\Phi^m Z_0) > W(\Phi^{2m} Z_0) > W(\Phi^{3m} Z_0) > W(Z_0),$$

which is impossible. No continuity or positivity assumption on W is needed. ∎

The same fixed potential therefore has different zero-output invariant sets in continuous and discrete time. Continuous output is the instantaneous velocity $u(t)$; numerical output is $(u_{n+1} + u_n)/2$. In the displayed orbit, averaging the nonlinear coupling over an entire sine period eliminates the force which would transfer motion to the damped coordinate. The physical energy still captures every discrete trajectory, and the explicit quadratic energy still attracts all phases in its invariant sublevel. The failure occurs in extending such strict convergence control to the whole phase space.

The numerical observability premise in [1, Theorem III.3], and the discrete invariant-set premise in [2, Proposition 1], remain essential qualifications of those results. Our example fails the numerical premise and hence does not refute either theorem. The argument makes no claim about a basin of attraction for the cycle or behavior under a change of the step. The accompanying technical note records the precise scope of the nearby sampled-data and energy-balance comparisons.

5 One fixed analytic potential at arbitrarily small steps

We now change the potential, keeping $D = \operatorname{diag}(1, 0)$ and the phase order $Z = (x, y, u, w)$. The objects in this section are distinct from those of Theorem 1. For $n \geq 1$, set

$$\begin{gathered} N_n = 4n, \qquad \theta_n = \frac{\pi}{2n}, \qquad h_n = \tan\frac{\pi}{4n}, \qquad A_n = n2^{n^2+4}, \\ b_{n,j} = A_n \cos(j\theta_n), \qquad 0 \leq j < n. \end{gathered} \tag{35}$$

Define the ordered product and potential by

$$\Pi(z) = \prod_{n=1}^{\infty} \prod_{j=0}^{n-1} \left(1 - \frac{z^2}{b_{n,j}^2}\right), \qquad F(z) = z^2 e^{-z^2} \Pi(z), \tag{36}$$

$$\mathcal{U}(x, y) = \frac{x^2}{2} + 2y^2 + \varepsilon \sin x \, F'(y), \qquad \varepsilon = 2^{-11}. \tag{37}$$

This is one fixed potential; it does not change when a step is chosen. The convergence of the product and its derivative bounds are proved below. Write

$$\mathcal{H}(q,v)=\mathcal{U}(q)+\frac{1}{2}\|v\|^2. \tag{38}$$

For any $h>0$ the literal AVF equations are

$$\begin{aligned} q^+-q &= h\frac{v^+ + v}{2},\\ v^+-v &= -h\int_0^1 \nabla\mathcal{U}(q+t(q^+-q))\,dt - hD\frac{v^+ + v}{2}. \end{aligned} \tag{39}$$

We denote the unique map established below by Ψ_h.

For a local energy common to $0<h\le 1$, put $\kappa=2^{-10}$, $d=4-\kappa^2$ and

$$P_*=\frac{1}{2\kappa^2 d}\begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14}\\ p_{12} & p_{22} & p_{23} & p_{24}\\ p_{13} & p_{23} & p_{33} & p_{34}\\ p_{14} & p_{24} & p_{34} & p_{44}\end{pmatrix}, \tag{40}$$

where

$$\begin{aligned} p_{11}&=\kappa^2(37-8\kappa^2), & p_{12}&=2\kappa(40-6\kappa^2-\kappa^4),\\ p_{13}&=\kappa^2(9-\kappa^2), & p_{14}&=-5\kappa,\\ p_{22}&=260-13\kappa^2-13\kappa^4, & p_{23}&=5\kappa d,\\ p_{24}&=-\kappa^2 d, & p_{33}&=\kappa^2(13-2\kappa^2),\\ p_{34}&=5\kappa(3-\kappa^2), & p_{44}&=65-7\kappa^2-2\kappa^4. \end{aligned} \tag{41}$$

Set

$$\begin{aligned} T_*&=\operatorname{tr}P_*=\frac{119113770161485}{2796202}, \qquad \ell_*=\frac{\det P_*}{T_*^3},\\ r_*^2&=\frac{1}{2^{22}(1+T_*)}, \qquad \eta_*=\frac{\ell_* r_*^2}{2}, \qquad \mathcal{V}(Z)=Z^T P_* Z. \end{aligned} \tag{42}$$

Theorem 2. *For the single potential* (37)*, the following statements hold.*

(i) *The product* (36) *converges uniformly on complex compact sets to an even entire function with* $\Pi(0)=1$*. The positive nodes are distinct and have no finite accumulation, and*

$$\sum_{n=1}^{\infty}\sum_{j=0}^{n-1} b_{n,j}^{-2}\le\frac{1}{576}, \qquad |F^{(a)}(y)|\le 64a! \quad (y\in\mathbb{R},\ 0\le a\le 5). \tag{43}$$

The zeros of F *are exactly* 0 *and* $\pm b_{n,j}$*, the former double and the others simple. The potential is real analytic, its unique critical point and minimizer is* 0*, and*

$$\frac{3}{4}I\preceq\nabla^2\mathcal{U}(q)\preceq\frac{17}{4}I \qquad (q\in\mathbb{R}^2). \tag{44}$$

(ii) *For every* $h>0$*, equations* (39) *define a unique smooth global map, with*

$$\frac{3}{8}\|Z\|^2\le\mathcal{H}(Z)\le\frac{17}{8}\|Z\|^2, \tag{45}$$

$$\mathcal{H}(\Psi_h Z) - \mathcal{H}(Z) = -h\left(\frac{u^+ + u}{2}\right)^2. \tag{46}$$

Every fixed-step orbit satisfies

$$\|Z_k\|^2 \le \frac{8}{3}\mathcal{H}(Z_0), \qquad \sum_{k=0}^{\infty} h\left(\frac{u_{k+1} + u_k}{2}\right)^2 \le \mathcal{H}(Z_0). \tag{47}$$

(iii) *The matrix P_* is positive definite and*

$$\ell_*\|Z\|^2 \le \mathcal{V}(Z) \le T_*\|Z\|^2. \tag{48}$$

For every $0 < h \le 1$ and every $\|Z\| \le r_*$,

$$\mathcal{V}(\Psi_h Z) - \mathcal{V}(Z) \le -\frac{h}{49}\|Z\|^2. \tag{49}$$

The same sublevel $\{\mathcal{V} \le \eta_*\}$ is invariant for each such map. For a fixed h in this interval, every orbit starting in that sublevel satisfies

$$\begin{gathered}\mathcal{V}(Z_k) \le \rho_h^k \mathcal{V}(Z_0), \qquad \|Z_k\|^2 \le \ell_*^{-1}\rho_h^k \mathcal{V}(Z_0), \\ \rho_h = 1 - \frac{h}{49(1 + T_*)} \in (0, 1).\end{gathered} \tag{50}$$

(iv) *The origin of the continuous system*

$$\dot{q} = v, \qquad \dot{v} = -\nabla\mathcal{U}(q) - Dv \tag{51}$$

is globally asymptotically stable.

(v) *For every $n \ge 1$, the actual map Ψ_{h_n} has the prime $4n$-cycle*

$$Z_{n,j} = (0, A_n\cos(j\theta_n), 0, -2A_n\sin(j\theta_n)), \quad 0 \le j < 4n, \tag{52}$$

in the displayed order. Its full initial velocity is zero, its damped midpoint output is zero on every edge, and its energy is $2A_n^2$. Its omega-limit set is exactly these $4n$ phases. Moreover,

$$h_n \longrightarrow 0, \qquad \min_{0 \le j < 4n} \|Z_{n,j}\| \ge A_n \longrightarrow \infty. \tag{53}$$

For every $n, m \ge 1$, no real-valued function W on $\mathbb{R}^4$ obeys

$$W(\Psi_{h_n}^m Z) < W(Z) \qquad \text{for all } Z \ne 0. \tag{54}$$

In particular, there is no $h_0 > 0$ such that every $0 < h < h_0$ and every initial phase produce convergence to the origin.

All quantifiers refer to the one fixed potential (37). The theorem gives a sequence of exceptional steps and escaping initial phases. It does not assert nonconvergence for every small step or on any fixed compact family of starts. The common local sublevel is nonempty and independent of h. Only the damped component of the midpoint velocity vanishes on the cycles; the full midpoint vector on the first edge is $(0, -A_n\sin\theta_n) \ne 0$.

6 Analytic bounds and global capture

Proof of Theorem 2(i). Concavity of sine on $[0, \pi/2]$ gives

$$2^{n^2+4} = \frac{A_n}{n} \le A_n \sin\frac{\pi}{2n} \le b_{n,j} \le A_n. \tag{55}$$

The nodes in each batch are positive and strictly decreasing. The minimum in batch $n+1$ exceeds the maximum in batch n, since $2^{2n+1} > n$. This proves distinctness and absence of finite accumulation. Consequently

$$S := \sum_{n,j} b_{n,j}^{-2} \le 2^{-8} \sum_{n\ge1} n 2^{-2n^2} \le 2^{-8} \sum_{n\ge1} n 4^{-n} = \frac{1}{576}.$$

For $|z| \le R$, the sum of $|z^2/b_{n,j}^2|$ is at most $R^2 S$. A finite product tail differs from 1 by at most $\exp(\sum |z^2/b_{n,j}^2|) - 1$, and every initial product has modulus at most $e^{R^2 S}$. Thus the ordered partial products are uniformly Cauchy on every disk. Their limit is entire, even, real on the real axis, and equal to 1 at the origin. Also

$$|\Pi(z)| \le e^{S|z|^2}. \tag{56}$$

Away from the nodes, sufficiently late factors have $|z^2/b_{n,j}^2| \le 1/2$. The power series for their logarithms has absolutely convergent sum, bounded termwise in absolute value by $2|z^2/b_{n,j}^2|$. The tail product is a nonzero exponential. Removing the one vanishing factor at a node proves that the remaining product is nonzero there. Thus the nonzero zeros are simple and no others occur. Since $\Pi(0) = 1$, the zero of F at 0 is exactly double and $F''(0) = 2$.

For $z = a + ib$ with $|b| \le 1$, inequality (56) gives

$$|F(a+ib)| \le (a^2+b^2) e^{-(1-S)a^2 + (1+S)b^2} \le (a^2+1) e^{-a^2/2 + 3/2} < 16. \tag{57}$$

Here $(t+1)e^{-t/2} \le 2$ for $t \ge 0$, and $e^{3/2} < 8$. A unit circle about any real number lies in this strip. Cauchy's formula gives $|F^{(a)}(y)| \le 16a! \le 64a!$. These are real-axis derivative bounds, with no claim that a nonconstant entire function is bounded on the whole complex plane.

The Hessian perturbation of $\mathrm{diag}(1, 4)$ is

$$\varepsilon \begin{pmatrix} -\sin x\, F'(y) & \cos x\, F''(y) \\ \cos x\, F''(y) & \sin x\, F'''(y) \end{pmatrix}.$$

Its entrywise absolute bounds are 1/32, 1/16 and 3/16. The maximum absolute row sum is at most 1/4, bounding its operator norm by symmetry. This proves (44). Since the value and gradient at the origin vanish, Taylor's formula gives

$$\frac{3}{8}\|q\|^2 \le \mathcal{U}(q) \le \frac{17}{8}\|q\|^2.$$

Strong convexity makes the origin the only critical point and minimizer. ∎

Proof of Theorem 2(ii). Adding kinetic energy gives (45). Fix any $h > 0$ and any phase. With $a = q^+ - q$, the original equations are equivalent to

$$\left(\frac{2}{h^2}I + \frac{1}{h}D\right)a + \int_0^1 \nabla\mathcal{U}(q+ta)\,dt = \frac{2}{h}v, \qquad v^+ = \frac{2}{h}a - v. \tag{58}$$

The first equation is the gradient equation for

$$J(a) = \frac{\|a\|^2}{h^2} + \frac{a^T Da}{2h} - \frac{2}{h}\langle v, a\rangle + \int_0^1 \frac{\mathcal{U}(q+ta) - \mathcal{U}(q)}{t}\,dt. \tag{59}$$

The singularity at $t = 0$ is removable as in (18). Differentiation gives

$$\nabla^2 J(a) = \frac{2}{h^2}I + \frac{1}{h}D + \int_0^1 t\nabla^2\mathcal{U}(q+ta)\,dt \succeq \left(\frac{2}{h^2} + \frac{3}{8}\right)I.$$

The function is coercive and strongly convex, hence has a unique critical point. Reconstruction in (58) proves existence and uniqueness for the original step. Invertibility of the same derivative and the implicit-function theorem give smooth local solution functions, which uniqueness joins to a global smooth map.

Set $\bar{v} = (v^+ + v)/2$. The chain rule along the actual position segment yields

$$\begin{aligned}\mathcal{H}(q^+, v^+) - \mathcal{H}(q, v) &= \left\langle \int_0^1 \nabla\mathcal{U}(q+ta)\,dt, a \right\rangle + \langle \bar{v}, v^+ - v\rangle \\ &= -h\bar{v}^T D\bar{v} = -h\left(\frac{u^+ + u}{2}\right)^2.\end{aligned}$$

Iteration of the global map and telescoping with $\mathcal{H} \geq 0$ prove (47). Before any local analysis, every actual step consequently obeys

$$\|\Psi_h Z\| \leq \sqrt{17/3}\,\|Z\| \leq 3\|Z\|. \tag{60}$$

∎

7 A strict local energy common to the small steps

The Hessian at the origin and continuous linearization are

$$K_* = \begin{pmatrix} 1 & \kappa \\ \kappa & 4 \end{pmatrix}, \qquad L_* = \begin{pmatrix} 0 & I \\ -K_* & -D \end{pmatrix}. \tag{61}$$

Direct polynomial multiplication gives

$$L_*^T P_* + P_* L_* = -I. \tag{62}$$

The leading principal minors of P_* at $\kappa = 2^{-10}$ are

$$\begin{aligned} &\frac{19398652}{4194303}, \qquad \frac{77341821357370550955}{1466015154176}, \\ &\frac{313450570486295166977}{4398045462528}, \qquad \frac{15961605287354846475164712 97}{4398045462528}. \end{aligned} \tag{63}$$

They are positive, so $P_* \succ 0$. The trace is the value in (42). Each eigenvalue is at most T_*, and their product is at most the smallest one times T_*^3. This proves (48). The accompanying exact program reproduces these identities.

Lemma 3. *The force remainder $R(q) = \nabla\mathcal{U}(q) - K_* q$ satisfies*

$$\|R(q)\| \le \frac{5}{2}\|q\|^3 \qquad (q \in \mathbb{R}^2). \tag{64}$$

Proof. Because F is even, $\mathcal{U}(-q) = \mathcal{U}(q)$ and $D^3\mathcal{U}(0) = 0$. Its fourth partial derivatives, in order of the number of y derivatives, are bounded in absolute value by

$$\frac{1}{32}, \quad \frac{1}{16}, \quad \frac{3}{16}, \quad \frac{3}{4}, \quad \frac{15}{4}.$$

For $a, b, c, d \in \mathbb{R}^2$, this implies

$$|D^4\mathcal{U}(q)[a,b,c,d]| \le \frac{15}{4}\|a\|_1\|b\|_1\|c\|_1\|d\|_1 \le 15\|a\|\|b\|\|c\|\|d\|.$$

The integral Taylor remainder for the gradient has coefficient $1/6$. Its constant, linear and quadratic terms are 0, $K_* q$ and 0. Hence the global remainder bound follows. ∎

Proof of Theorem 2(iii). We prove the decrement on the full input ball before invariance. Let $0 < h \le 1$, $\|Z\| \le r_*$, and $Z^+ = \Psi_h Z$. Put $\xi = (Z + Z^+)/2$. The literal equations give

$$Z^+ - Z = h(L_*\xi + e), \qquad e = -\begin{pmatrix} 0 \\ \displaystyle\int_0^1 R(q + t(q^+ - q))\, dt \end{pmatrix}. \tag{65}$$

The global bound (60) places every point of the position segment inside the ball of radius $3\|Z\|$. Therefore

$$\|e\| \le \frac{135}{2}\|Z\|^3 \le 128\|Z\|^3, \qquad \|\xi\| \le 2\|Z\|. \tag{66}$$

Also $\|L_*\| \le \|L_*\|_F = \sqrt{20 + 2\kappa^2} < 5$. Since $128 r_*^2 \le 1/4$, rearranging (65) gives

$$\|Z\| \le \left(1 + \frac{h}{2}\|L_*\|\right)\|\xi\| + \frac{h}{2}\|e\| \le \frac{7}{2}\|\xi\| + \frac{1}{8}\|Z\|.$$

Hence $\|\xi\| \ge \|Z\|/4$. The exact quadratic difference and (62) now imply

$$\begin{aligned} \mathcal{V}(Z^+) - \mathcal{V}(Z) &= 2h\xi^T P_*(L_*\xi + e) \\ &= -h\|\xi\|^2 + 2h\xi^T P_* e \\ &\le -h\left(\frac{1}{16} - 512 T_* r_*^2\right)\|Z\|^2 \le -\frac{h}{49}\|Z\|^2. \end{aligned} \tag{67}$$

The last inequality uses $512T_*r_*^2 < 1/8192$ and $511/8192 > 1/49$. This proves the input-ball decrement, including zero, without assuming a local bound on the successor.

If $\mathcal{V}(Z) \le \eta_*$, coercivity gives $\|Z\|^2 \le r_*^2/2$. The decrement just proved gives $\mathcal{V}(\Psi_h Z) \le \mathcal{V}(Z) \le \eta_*$, so the common sublevel is invariant. For a fixed h, within it,

$$\mathcal{V}(Z_{k+1}) \le \mathcal{V}(Z_k) - \frac{h}{49}\|Z_k\|^2 \le \left(1 - \frac{h}{49(1+T_*)}\right)\mathcal{V}(Z_k).$$

The factor lies in $(0,1)$. Iteration and the lower spectral bound give (50). ∎

The function and trapping sublevel are common to the entire step interval. The convergence estimate is stated for each fixed step; no claim about convergence for arbitrary variable steps is needed.

8 Continuous convergence and cold orbits

Proof of Theorem 2(iv). Bounded Hessian makes the continuous vector field globally Lipschitz. Every solution exists globally and obeys

$$\frac{d}{dt}\mathcal{H}(q(t), v(t)) = -u(t)^2.$$

The coercive bounds give compact capture and Lyapunov stability. If a trajectory stays in $u = 0$, then $x = c$ is constant and

$$c + \varepsilon \cos c\, F'(y(t)) = 0. \tag{68}$$

The case $\cos c = 0$ would force $c = 0$, a contradiction. Otherwise $F'(y(t))$ is constant. If $y(t)$ were nonconstant, its continuous image over some time interval would contain a nondegenerate real interval. The analytic identity theorem would then make F' constant everywhere, contradicting $F''(0) = 2$. Thus both positions are constant and both velocities vanish. The acceleration equations imply $\nabla\mathcal{U}(c, y) = 0$, so the state is the origin.

The energy of any bounded trajectory has a limit. By continuous dependence applied to convergent time translates, every omega-limit point generates a forward trajectory of that constant energy. Such a trajectory stays in $u = 0$ and is therefore zero. Compactness implies convergence to the origin; together with Lyapunov stability this proves global asymptotic stability. ∎

Proof of Theorem 2(v). Fix n and abbreviate $A = A_n$, $\theta = \theta_n$, $h = h_n$. The scalar positions in (52) belong to $\{0\} \cup \{\pm b_{n,j} : 0 \le j < n\}$, so F vanishes at each one. Although scalar positions repeat, consecutive ones differ:

$$y_{j+1} - y_j = -2A \sin\big((2j+1)\theta/2\big) \sin(\theta/2) \neq 0. \tag{69}$$

Indeed equality would make the odd integer $2j + 1$ a multiple of $4n$. This includes the closing edge.

On every position segment, $x = 0$, $\mathcal{U}_x(0, y) = \varepsilon F'(y)$ and $\mathcal{U}_y(0, y) = 4y$. The first averaged force is exactly

$$\varepsilon \int_0^1 F'(y_j + t(y_{j+1} - y_j))\, dt = \varepsilon \frac{F(y_{j+1}) - F(y_j)}{y_{j+1} - y_j} = 0. \tag{70}$$

The denominator is nonzero even when one endpoint is zero. The second averaged force is $2(y_j + y_{j+1})$. The remaining equations are

$$y_{j+1} - y_j = h\frac{w_{j+1} + w_j}{2}, \qquad w_{j+1} - w_j = -2h(y_{j+1} + y_j). \tag{71}$$

They follow from the angle addition formulas and $h = \tan(\theta/2)$. Equivalently, their matrix on (y, w) is

$$\frac{1}{1 + h^2} \begin{pmatrix} 1 - h^2 & h \\ -4h & 1 - h^2 \end{pmatrix},$$

which is rotation by $-\theta$ on $(y, w/2)$. All four coordinates of each proposed edge solve (39). Global uniqueness makes them the actual map values.

Equality of two phases implies equality of both sine and cosine, so their indices differ by a multiple of $4n$. Thus the least period is $4n$. The full initial velocity is zero. The full midpoint velocity on every edge is nonzero by (69) and (71); only its damped first coordinate vanishes. Direct calculation yields

$$\begin{aligned} &\mathcal{H}(Z_{n,j}) = 2A_n^2, \\ &\|Z_{n,j}\|^2 = A_n^2\big(\cos^2(j\theta_n) + 4\sin^2(j\theta_n)\big) \geq A_n^2. \end{aligned} \tag{72}$$

All $4n$ phases recur and no others occur, proving the omega-limit claim. The limits $h_n \to 0$ and $A_n \to \infty$ give (53).

For any $m \geq 1$, the m-step map permutes these nonzero phases and returns to each after $4n/\gcd(4n, m)$ applications. Iterating (54) that many times would give $W(Z) < W(Z)$, which is impossible. For any proposed $h_0 > 0$, choose n with $h_n < h_0$ and start at $Z_{n,0}$. This periodic orbit cannot converge to the origin. ∎

This disproves a single positive threshold valid for every initial phase of the fixed potential. It does not disprove a statement with the other order of quantifiers: for each fixed compact set of starts there may be a sufficiently small threshold depending on that set. The witnesses escape every compact phase set and every fixed energy sublevel. Their absence from a compact set is not a proof of convergence there; the positive result proved here is the common local sublevel in Theorem 2(iii).

The distinction is consistent with the inspected general literature. Kloeden and Lorenz [5, Theorem 1.1] obtain nearby attracting sets under uniform one-step hypotheses, without identifying all motion inside them as convergence to an equilibrium. Karafyllis and Grüne [6, Theorem 4.12 and Remark 4.13] add local exponential stability and a finite-stage Runge–Kutta framework, with state- or compact-dependent step control. Their theorem is not quoted here as an exact-convergence result for a nonpolynomial AVF integral.

These comparisons, and the numerical-observability and discrete invariant-set hypotheses in [1, 2], limit the scope. The method, energy identities, analytic-product estimates and local Lyapunov tools are inherited. No firstness or complete-priority conclusion is drawn.

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